

## Section B

8(a)(i) The loss in GPE of the ball is converted into its KE.

$$mgh = (1/2)mv^2$$

$$v = \sqrt{2gh} = \sqrt{2(9.81)(0.70)}$$

$$= 3.706 \text{ m s}^{-1} \text{ (For 'Show' questions, must show substitution and unrounded answer first)}$$

$$= 3.7 \text{ m s}^{-1}$$

(ii)

Tension,  $T$



Weight,  $W$

(iii) Centripetal force,  $F_c = \frac{mv^2}{r} = \frac{(0.050)(3.706)^2}{(1.50)}$   
 $= 0.4578 \text{ N} = 0.458 \text{ N}$

(iv) The ball is moving in a circular path, so it is experiencing an acceleration. According to Newton's first law, there must be a resultant force on the ball. So  $T \neq W$ , otherwise there will be no resultant force on the ball.

(v) Applying Newton's second law,  
 $F_{\text{net}} = ma = F_c$   
 $T - W = F_c$   
 $T = W + F_c = (0.050)(9.81) + 0.4578$   
 $= 0.9483 = 0.948 \text{ N}$

(b)(i) The gravitational field strength at a point is the gravitational force per unit mass acting at that point.

(ii)  $g = \frac{GM}{d^2}$ , where  $d$  is the distance between the COMs of Charon and Pluto  
 $g = \frac{(6.67 \times 10^{-11})(1.27 \times 10^{22})}{(1.96 \times 10^7)^2}$   
 $= 2.205 \times 10^{-3} \text{ m s}^{-2}$   
 $= 2.21 \times 10^{-3} \text{ m s}^{-2}$

$g$  points towards the centre of Pluto.

(iii)  $\omega$  is the angular speed about the joint centre of mass.  
 Centripetal force on pluto,  $F_P = m_P r_P \omega^2$  & centripetal force on Charon,  $F_C = m_C r_C \omega^2$

The centripetal force is provided by the gravitational force on each body, and the gravitational forces have the same magnitude, according to Newton's third law.

So  $m_P r_P \omega^2 = m_C r_C \omega^2$   
 $\therefore \frac{m_P}{m_C} = \frac{r_C}{r_P}$

(iv)  $r_P = \left(\frac{m_C}{m_P}\right) r_C$   
 $r_P + r_C = 1.96 \times 10^7 \text{ m}$   
 $\left(\frac{m_C}{m_P}\right) r_C + r_C = 1.96 \times 10^7 \text{ m}$   
 $r_C = \frac{1.96 \times 10^7 \text{ m}}{1 + \left(\frac{m_C}{m_P}\right)} = \frac{1.96 \times 10^7 \text{ m}}{1 + \left(\frac{1.50 \times 10^{21} \text{ kg}}{1.27 \times 10^{22} \text{ kg}}\right)}$   
 $= 1.753 \times 10^7 \text{ m} = 1.75 \times 10^7 \text{ m}$

